Lesson 1.3: Introduction to Matrices

Learning Outcomes: At the end of the lesson, the students are able to:

1. perform basic matrix operations.

2. establish properties of matrix operations.

Basic Matrix Operations

In this lesson, we shall see that a matrix is a mathematical object consisting of vectors as its *columns*. We shall also learn the basic matrix operations and their properties.

Definition 31

Let m and n be positive integers. An m-by-n **matrix** (or a matrix of size m-by-n) is a rectangular array of scalars arranged in m rows and n columns usually expressed as

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

The element a_{ij} is called as the **(i,j)-entry** of the matrix. The matrix above is abbreviated as $[a_{ij}]$, when its size is clear from the context. Two matrices are **equal** if they have the same size and corresponding entries are equal.

Note that we identify a matrix with only one column as a vector called a **column vector** and call a matrix with only one row as **row vector**. We also identify a 1-by-1 matrix as a scalar.

Example 32. Consider the matrix

$$\begin{bmatrix} 3 & -2 & \sqrt{5} & 0.34 \\ -1 & -\pi & 4 & 2/3 \\ 0 & 2.5 & -98 & 1 \end{bmatrix}.$$

The given matrix is a 3-by-4 matrix. The second row of this matrix is the row vector

$$\begin{bmatrix} -1 & -\pi & 4 & 2/3 \end{bmatrix}$$

and its third column is the column vector

$$\begin{bmatrix} \sqrt{5} \\ 4 \\ -98 \end{bmatrix}.$$

The $(2,3)$-entry of the given matrix is 4. Note that the columns of the given matrices may be identified as vectors in $\mathbb{R}^3$, while its rows may be identified as vectors in $\mathbb{R}^4$.

Example 33. The following table shows the average total units sold of products A,B, and C in two different branches of store X in a given day.

	branch 1	branch 2
product A	34	45
product B	52	67
product C	93	78

Note that such a table is a real life example of a 3-by-2 matrix

$$\begin{bmatrix} 34 & 45 \\ 52 & 67 \\ 93 & 78 \end{bmatrix}.$$

The third row represents the number of units of product C sold in both the branches, while the second column represents the number of units sold in branch 2 of the three different products.

Definition 34: Basic Matrix Operations

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be m-by-n matrices.

- The **sum** of A and B is the matrix $A + B := [a_{ij} + b_{ij}]$.
- The **scalar multiple** of A by a scalar c is the m-by-n matrix $cA := [ca_{ij}]$.

- The **zero matrix** $0_{m,n}$ is the m-by-n matrix whose entries are all zeroes.
- We also define $-A := [-a_{ij}]$ and $A - B := A + (-B)$.

The definition above tells us that matrix addition and scalar multiple is similar to that of the corresponding vector operations. We add matrices of the same size entry-wise and we multiply a scalar to a matrix by multiplying the scalar to every entry of the matrix.

Example 35. Given the matrices

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 1 & 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 4 & 3 & 0 \\ 1 & -5 & 5 \end{bmatrix}, \text{ and } C = \begin{bmatrix} 2 & 1 & -1 \\ 6 & -1 & 2 \end{bmatrix}.$$

Calculate $2A - 3B + C$.

Solution. Note that

- $2A = \begin{bmatrix} 2 & 6 & -4 \\ 2 & 0 & 4 \end{bmatrix}$, and
- $3B = \begin{bmatrix} 12 & 9 & 0 \\ 3 & -5 & 5 \end{bmatrix}$.

Hence,

$$\begin{aligned} 2A - 3B + C &= \begin{bmatrix} 2 & 6 & -4 \\ 2 & 0 & 4 \end{bmatrix} - \begin{bmatrix} 12 & 9 & 0 \\ 3 & -5 & 5 \end{bmatrix} + \begin{bmatrix} 2 & 1 & -1 \\ 6 & -1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} -8 & -2 & -5 \\ 5 & 4 & 1 \end{bmatrix}. \end{aligned}$$

Example 36. Not every pair of matrices may be added together. For example, the sum

$$\begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix} + \begin{bmatrix} 3 & 4 & -1 & 2 \\ 1 & 1 & -5 & 1 \end{bmatrix}$$

is *undefined* since the given matrices are of different sizes.

Since matrix addition and scalar multiplication is defined similarly as its vector operation counterparts, it is not surprising that the usual properties also hold. We state them and omit the proofs.

Theorem 37

Let A, B, and C be m-by-n matrices and let r and s be scalars. Then

1. $A + B = B + A$.
2. $A + (B + C) = (A + B) + C$.
3. $A + 0_{m,n} = A$.
4. $A + (-A) = 0_{m,n}$.
5. $r(A + B) = rA + rB$.
6. $(r + s)A = rA + sB$.
7. $(rs)A = r(sA)$.
8. $1A = A$.

We are now ready to introduce matrix multiplication. At first glance, the definition may not seem intuitive, but as we shall see later, it is quite natural.

Definition 38: Matrix Multiplication

Let $A = [a_{ij}]$ be an m-by-p matrix and $B = [b_{ij}]$ be a p-by-n matrix. The **product** of A and B is defined as the m-by-n matrix $AB = [c_{ij}]$, where

$$\begin{aligned} c_{ij} &= \sum_{k=1}^{p} a_{ik} b_{kj} \\ &= a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ip}b_{pj}, \end{aligned}$$

for $i = 1, 2, \ldots, m$ and $j = 1, 2, \ldots, n$.

Notice that the definition for matrix multiplication is not as expected as matrix addition and scalar multiplication. As we shall reveal in the future lessons, this definition is not the most straightforward one, but it is the most natural. To explain the definition, we first look at the size requirements.

$$\begin{array}{ccccc} A & \times & B & = & AB \\ (m\text{-by-}p) & & (p\text{-by-}n) & & (m\text{-by-}n) \end{array}$$

The number of columns of A, which is p, equals the number of rows of B. This ensures that the rows of A and the columns of B are in the same space $\mathbb{R}^p$. This must be required because in order to obtain the (i,j)-entry of the product AB, we must take the dot product of row i of A and the column j of B, that is,

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ip}b_{pj}.$$

To further illustrate how matrix multiplication works, we setup the following table.

$$\begin{array}{cccc|cccccc}
 & & & & b_{11} & b_{12} & \cdots & b_{1j} & \cdots & b_{1n} \\
 & & & & b_{21} & b_{22} & \cdots & b_{2j} & \cdots & b_{2n} \\
 & & & & \vdots & \vdots & & \vdots & & \vdots \\
 & & & & b_{p1} & b_{p2} & \cdots & b_{pj} & \cdots & b_{pn} \\
\hline
a_{11} & a_{12} & \cdots & a_{1p} & c_{11} & c_{12} & \cdots & c_{1j} & \cdots & c_{1n} \\
a_{21} & a_{22} & \cdots & a_{2p} & c_{21} & c_{22} & \cdots & c_{2j} & \cdots & c_{2n} \\
\vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots & & \vdots \\
a_{i1} & a_{i2} & \cdots & a_{ip} & c_{i1} & c_{i2} & \cdots & c_{ij} & \cdots & c_{in} \\
\vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots & & \vdots \\
a_{m1} & a_{m2} & \cdots & a_{mp} & c_{m1} & c_{m2} & \cdots & c_{mj} & \cdots & c_{mn}
\end{array}$$

$$(\text{row } i \text{ of } A) \cdot (\text{column } j \text{ of } B) = c_{ij}$$

Example 39. Let $A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -3 & 1 \\ 0 & 1 & 2 \end{bmatrix}$. Since A is 2-by-2 and B is 2-by-3, the product AB is defined and its size is 2-by-3. For instance, to get the $(1,3)$-entry of AB, we take the dot product of the first row of A and the third column of B.

$$(1,3)\text{-entry of } AB = 2(1) + (-1)(2) = 0.$$

We do the same method for each of the (i, j)-entry of AB, $i = 1, 2$ and $j = 1, 2, 3$. So,

$$
\begin{aligned}
AB &= \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & -3 & 1 \\ 0 & 1 & 2 \end{bmatrix} \\
&= \begin{bmatrix} 2(2) + (-1)(0) & 2(-3) + (-1)(1) & 2(1) + (-1)(2) \\ 1(2) + 0(0) & 1(-3) + 0(1) & 1(1) + 0(2) \end{bmatrix} \\
&= \begin{bmatrix} 4 & -7 & 0 \\ 2 & -3 & 1 \end{bmatrix}.
\end{aligned}
$$

We can also calculate the product AB using the tabular method shown above.

$$
\begin{array}{cc|ccc}
\multicolumn{2}{c|}{\multirow{2}{*}{AB}} & 2 & -3 & 1 \\
 & & 0 & 1 & 2 \\
\hline
2 & -1 & 4 & -7 & 0 \\
1 & 0 & 2 & -3 & 1
\end{array}
$$

(row 1) $\cdot$ (column 3) $=$ $(1, 3)$-entry

Observe that the product BA is **not** defined since B has 3 columns while A has 2 rows and $3 \neq 2$.

Remark. The product BA may not be defined even when AB is defined.

Example 40. For each of the following pair of matrices, compute the products AB and BA, whenever possible.

1. $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 2 & -2 \\ 4 & 3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 & -3 \\ 1 & 0 & 2 \\ -2 & 5 & 1 \end{bmatrix}$

2. $A = \begin{bmatrix} 0 & 2 & -3 \\ 4 & -1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ 1 & 4 \end{bmatrix}$

Solutions.

1. Multiplying AB, we get

$$\begin{array}{ccc|ccc} & & & 2 & -1 & -3 \\ & & & 1 & 0 & 2 \\ & & & -2 & 5 & 1 \\ \hline 1 & 2 & -1 & 6 & -6 & 0 \\ 0 & 2 & -2 & 6 & -10 & 2 \\ 4 & 3 & 1 & 9 & 1 & -5 \end{array} \implies AB = \begin{bmatrix} 6 & -6 & 0 \\ 6 & -10 & 2 \\ 9 & 1 & -5 \end{bmatrix}.$$

Now, multiplying BA, we get

$$\begin{array}{ccc|ccc} & & & 1 & 2 & -1 \\ & & & 0 & 2 & -2 \\ & & & 4 & 3 & 1 \\ \hline 2 & -1 & -3 & -10 & -7 & -3 \\ 1 & 0 & 2 & 9 & 8 & 1 \\ -2 & 5 & 1 & 2 & 9 & -7 \end{array} \implies BA = \begin{bmatrix} -10 & -7 & -3 \\ 9 & 8 & 1 \\ 2 & 9 & -7 \end{bmatrix}.$$

2. Multiplying AB, we get

$$\begin{array}{ccc|cc} & & & 1 & 1 \\ & & & 2 & -1 \\ & & & 1 & 4 \\ \hline 0 & 2 & -3 & 1 & -14 \\ 4 & -1 & 1 & 3 & 9 \end{array} \implies AB = \begin{bmatrix} 1 & -14 \\ 3 & 9 \end{bmatrix}.$$

Now, multiplying BA, we get

$$\begin{array}{cc|ccc} & & 0 & 2 & -3 \\ & & 4 & -1 & 1 \\ \hline 1 & 1 & 4 & 1 & -2 \\ 2 & -1 & -4 & 5 & -7 \\ 1 & 4 & 16 & -2 & 1 \end{array} \implies BA = \begin{bmatrix} 4 & 1 & -2 \\ -4 & 5 & -7 \\ 16 & -2 & 1 \end{bmatrix}.$$

As the previous example shows, even when both AB and BA are defined, they are not necessarily equal. This is why we remark the following.

Matrix multiplication is **not** commutative.

Example 41. Consider the following table from **Example 33**.

	branch 1	branch 2
product A	34	45
product B	52	67
product C	93	78

Suppose that the unit prices of products A, B, and C are PhP 40.00, PhP 70.00, and PhP 60.00, respectively and that shoppers gets a 10% discount on holidays. The table below shows the matrix of unit prices for each of the products under the regular price and discounted price.

in PhP	product A	product B	product C
regular price	40	70	60
discounted price	36	63	54

If we let $Q = \begin{bmatrix} 34 & 45 \\ 52 & 67 \\ 93 & 78 \end{bmatrix}$ be the quantity sold matrix and $P = \begin{bmatrix} 40 & 70 & 60 \\ 36 & 63 & 54 \end{bmatrix}$ be the unit price matrix, then the product PQ can be obtained as follows.

				branch 1	branch 2
				34	45
				52	67
				93	78
regular price	40	70	60	10,580	11,170
discounted price	36	63	54	9,522	10,053

Hence, $PQ = \begin{bmatrix} 10,580 & 11,170 \\ 9,522 & 10,053 \end{bmatrix}$. This matrix shows the total sales of both branches under the two pricing schemes: regular and discounted. In tabular form, we may present PQ as follows.

sales in PhP	branch 1	branch 2
regular price	10,580	11,170
discounted price	9,522	10,053

Here, we can conclude that if it were a regular day, the total sales of branch 1 is PhP 10,580, while that of branch 2 is PhP 11,170. On the other hand, if it were a holiday, branch 1 has a total sales of PhP 9,522 and branch 2 has PhP 10,053.

Observation. The jth column of the product AB is equal to $A\vec{c_j}$, where $\vec{c_j}$ is the jth column of B. If we write

$$B = \begin{bmatrix} \vec{c_1} & \vec{c_2} & \cdots & \vec{c_n} \end{bmatrix},$$

then

$$AB = \begin{bmatrix} A\vec{c_1} & A\vec{c_2} & \cdots & A\vec{c_n} \end{bmatrix}$$

To illustrate this observation, let $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$. Let $\vec{c_1} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\vec{c_2} = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$, and $\vec{c_3} = \begin{bmatrix} 0 \\ 4 \end{bmatrix}$. Since

- $A\vec{c_1} = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$;
- $A\vec{c_2} = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} -3 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 10 \end{bmatrix}$; and
- $A\vec{c_3} = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 4 \end{bmatrix} = \begin{bmatrix} 8 \\ 12 \end{bmatrix}$,

it follows that $AB = \begin{bmatrix} 5 & -1 & 8 \\ 5 & 10 & 12 \end{bmatrix}$.

Example 42. Let $A = \begin{bmatrix} 1 & 4 \\ -1 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$, and $C = \begin{bmatrix} -3 & -1 \\ 2 & 0 \end{bmatrix}$. Calculate the products $(AB)C$ and $A(BC)$.

Solution. For the product $(AB)C$, we first perform AB and multiply the result to C.

Step 1: Setup

$$\begin{array}{cc|cc|cc} & & 2 & 1 & -3 & -1 \\ & & 0 & 3 & 2 & 0 \\ \hline 1 & 4 & & & & \\ -1 & -1 & & & & \end{array}$$

Step 2: AB

$$\begin{array}{cc|cc|cc} & & 2 & 1 & -3 & -1 \\ & & 0 & 3 & 2 & 0 \\ \hline 1 & 4 & 2 & 13 & & \\ -1 & -1 & -2 & -4 & & \end{array}$$

Step 3: $(AB)C$

$$\begin{array}{cc|cc|cc} & & 2 & 1 & -3 & -1 \\ & & 0 & 3 & 2 & 0 \\ \hline 1 & 4 & 2 & 13 & 20 & -2 \\ -1 & -1 & -2 & -4 & 2 & 2 \end{array}$$

Therefore, $(AB)C = \begin{bmatrix} 20 & -2 \\ 2 & 2 \end{bmatrix}$.

On the other hand, for the product $A(BC)$, we first perform BC and multiply A to the result.

Step 1: Setup

$$\begin{array}{cc|cc} & & -3 & -1 \\ & & 2 & 0 \\ \hline 2 & 1 & & \\ 0 & 3 & & \\ \hline 1 & 4 & & \\ -1 & -1 & & \end{array}$$

Step 2: BC

$$\begin{array}{cc|cc} & & -3 & -1 \\ & & 2 & 0 \\ \hline 2 & 1 & -4 & -2 \\ 0 & 3 & 6 & 0 \\ \hline 1 & 4 & & \\ -1 & -1 & & \end{array}$$

Step 3: $A(BC)$

$$\begin{array}{cc|cc} & & -3 & -1 \\ & & 2 & 0 \\ \hline 2 & 1 & -4 & -2 \\ 0 & 3 & 6 & 0 \\ \hline 1 & 4 & 20 & -2 \\ -1 & -1 & -2 & 2 \end{array}$$

Therefore, $A(BC) = \begin{bmatrix} 20 & -1 \\ 1 & 1 \end{bmatrix}$.

Notice that in the previous example, $(AB)C = A(BC)$.

Theorem 43: Associativity of Matrix Multiplication

If the product $(AB)C$ is defined, then so is the product $A(BC)$ and that $(AB)C = A(BC)$.

Proof. Let m, n, p, q be given positive integers and let $A = [a_{ij}]$ be an m-by-p matrix, $B = [b_{ij}]$ be a p-by-n matrix, and $C = [c_{ij}]$ be a q-by-n matrix. If $AB = [d_{ij}]$, then AB is m-by-q and for all i and j,

$$d_{ij} = \sum_{k=1}^{p} a_{ik} b_{kj}. \tag{1.1}$$

If $(AB)C = [e_{ij}]$, then $(AB)C$ is m-by-n and for all i and j,

$$e_{ij} = \sum_{\ell=1}^{q} d_{i\ell} c_{\ell j}. \tag{1.2}$$

On the other hand, let $BC = [f_{ij}]$. Then BC is p-by-n and for all i and j,

$$f_{ij} = \sum_{\ell=1}^{q} b_{i\ell} c_{\ell j}. \tag{1.3}$$

Morever, if $A(BC) = [g_{ij}]$, then $A(BC)$ is m-by-n and for all i and j,

$$\begin{aligned} g_{ij} &= \sum_{k=1}^{p} a_{ik} f_{kj} = \sum_{k=1}^{p} \left[a_{ik} \left(\sum_{\ell=1}^{q} b_{k\ell} c_{\ell j} \right) \right] \\ &= \sum_{\ell=1}^{q} \left[\left(\sum_{k=1}^{p} a_{ik} b_{k\ell} \right) c_{\ell j} \right] = \sum_{\ell=1}^{q} d_{i\ell} c_{\ell j} \\ &= e_{ij}. \end{aligned} \tag{1.4}$$

Therefore, $A(BC) = [g_{ij}] = [e_{ij}] = (AB)C$.

The following properties may be proven analogously and is left for students to verify as an exercise.

Theorem 44

Let A, B, and C be matrices such that the products below are defined. Then

1. $A(B + C) = AB + AC$.
2. $(A + B)C = AC + BC$.
3. $(rA)(sB) = (rs)(AB)$, for all scalars r and s.

With associativity in mind, we may leave the expression ABC without grouping symbols without a possible confusion.

If A is an m-by-n matrix, notice that the product AA may only be defined in the case when $m = n$, that is, when the number of rows and number of columns of A are equal. In this case, we say that A is a **square matrix**.

Definition 45: Exponents

For a square matrix A, we define $A^1 := A$ and for an integer $k \geq 2$, we define $A^k := A^{k-1}A$.

As an example, it follows that

$$A^4 = A^3A = (A^2A)A = ((A^1A)A)A = ((AA)A)A,$$

or simply $A^4 = AAAA$ by associative property.

Example 46. Calculate A^4, if $A = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$.

Solution. From **Theorem 44**,

$$A^4 = \left(\tfrac{1}{\sqrt{2}}\right)^4 \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}^4 = \tfrac{1}{4}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}^4.$$

To calculate $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}^4$, we perform the following multiplication:

$$\begin{array}{cc|cc|cc|cc} & & 1 & -1 & 1 & -1 & 1 & -1 \\ & & 1 & 1 & 1 & 1 & 1 & 1 \\ \hline 1 & -1 & 0 & -2 & -2 & -2 & -4 & 0 \\ 1 & 1 & 2 & 0 & 2 & -2 & 0 & -4 \end{array}$$

which means $A^4 = \frac{1}{4}\begin{bmatrix} -4 & 0 \\ 0 & -4 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$. ▨

We now introduce an important matrix known as the **identity matrix**.

Definition 47: Identity Matrix

The n-by-n **identity matrix** is the n-by-n matrix I_n whose (i,j)-entry is

$$\delta_{ij} = \begin{cases} 1, & \text{if } i = j; \\ 0, & \text{if } i \neq j. \end{cases}$$

Note that

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \ldots$$

Notice that the columns of I_n are the *standard unit vectors* $\vec{e_1}, \vec{e_2}, \ldots, \vec{e_n}$, in this order. The entries are all zeroes except for 1's along the **main diagonal** (entries where $i = j$.)

As we shall see in the next theorem, the identity matrix is quite a generalization of the identity element 1 for multiplication of real numbers.

Theorem 48

Let A be an m-by-n matrix. Then

$$AI_n = A \text{ and } I_m A = A.$$

Proof. We shall only prove the first equality and leave the proof of the other to the student.

Let $A = [a_{ij}]$ be m-by-n matrix. If b_{ij} is the (i,j)-entry of AI_n, then by definition of matrix multiplication,

$$b_{ij} = \sum_{k=1}^{n} a_{ik}\delta_{kj},$$

for all i and j, where $\delta_{kj} = 1$, for $k = j$ and $\delta_{kj} = 0$, otherwise. As k runs from 1 to n, we see that at a certain point, k hits the value of j. Thus,

$$\begin{aligned} b_{ij} &= \sum_{k=1}^{n} a_{ik}\delta_{kj} \\ &= a_{i1}\delta_{1j} + a_{i2}\delta_{2j} + \cdots + a_{ij}\delta_{jj} + \cdots + a_{in}\delta_{nj} \\ &= 0 + 0 + \cdots + a_{ij}(1) + \cdots + 0 \\ &= a_{ij}. \end{aligned}$$

Therefore, $AI_n = A$.

Definition 49: Transpose of a Matrix

The **transpose** of an m-by-n matrix A is the n-by-m matrix A^t whose (i,j)-entry is the $(j,1)$-entry of A.

As a direct consequence of the definition, the rows of A^t are precisely the columns of A and that the columns of A^t are the rows of A, in order.

Example 50. Let $A = \begin{bmatrix} 1 & 2 & -1 & 2 \\ 0 & -3 & 1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 4 & 0 & 1 \\ 2 & -5 & 3 & 2 \end{bmatrix}$. Whenever possible, calculate $(BA^t + AB^t)^t$.

Solution.

- $A^t = \begin{bmatrix} 1 & 0 \\ 2 & -3 \\ -1 & 1 \\ 2 & 3 \end{bmatrix}$.
- BA^t is defined.

$$\begin{array}{cccc|cc} & & & & 1 & 0 \\ & & & & 2 & -3 \\ & & & & -1 & 1 \\ & & & & 2 & 3 \\ \hline -1 & 4 & 0 & 1 & 9 & -9 \\ 2 & -5 & 3 & 2 & -7 & 24 \end{array} \implies AB^t = \begin{bmatrix} 9 & -9 \\ -7 & 24 \end{bmatrix}.$$

- $B^t = \begin{bmatrix} -1 & 2 \\ 4 & -5 \\ 0 & 3 \\ 1 & 2 \end{bmatrix}$.

- AB^t is defined.

$$\begin{array}{cccc|cc} & & & & -1 & 2 \\ & & & & 4 & -5 \\ & & & & 0 & 3 \\ & & & & 1 & 2 \\ \hline 1 & 2 & -1 & 2 & 7 & -7 \\ 0 & -3 & 1 & 3 & -9 & 24 \end{array} \implies AB^t = \begin{bmatrix} 7 & -7 \\ -9 & 24 \end{bmatrix}.$$

- $BA^t + AB^t = \begin{bmatrix} 9 & -9 \\ -7 & 24 \end{bmatrix} + \begin{bmatrix} 7 & -7 \\ -9 & 24 \end{bmatrix} = \begin{bmatrix} 16 & -16 \\ -16 & 24 \end{bmatrix}$.

- Therefore, $(BA^t + AB^t)^t = \begin{bmatrix} 16 & -16 \\ -16 & 24 \end{bmatrix}^t = \begin{bmatrix} 16 & -16 \\ -16 & 24 \end{bmatrix}$.

▨

The following are the properties of matrix transpose.

Theorem 51

Let A and B be matrices such that the expressions below are defined. Then

1. $(A^t)^t = A$.
2. $(A + B)^t = A^t + B^t$.
3. $(cA)^t = cA^t$.
4. $(AB)^t = B^t A^t$.

Proof. The first three properties are self-explanatory. We only provide a proof for (4). Note that

$$\begin{aligned} (i,j)\text{-entry of } (AB)^t &= (j,i)\text{-entry of } AB \\ &= (\text{row } j \text{ of } A) \cdot (\text{column } i \text{ of } B) \\ &= (\text{row } i \text{ of } B^t) \cdot (\text{column } j \text{ of } A^t) \\ &= (i,j)\text{-entry of } B^t A^t. \end{aligned}$$

Therefore, $(AB)^t = B^t A^t$.

Inductively, if $A_1, A_2, \ldots, A_k$ are matrices of appropriate sizes, then

$$(A_1 A_2 \cdots A_k)^t = A_k^t A_{k-1}^t \cdots A_1^t.$$

Example 52. In **Example 50**, note that the matrices $(BA^t + AB^t)^t$ and $BA^t + AB^t$ are actually equal. This is not surprising since by **Theorem 51**,

$$\begin{aligned}(BA^t + AB^t)^t &= (BA^t)^t + (AB^t)^t \\ &= (A^t)^t B^t + (B^t)^t A^t \\ &= AB^t + BA^t \\ &= BA^t + AB^t.\end{aligned}$$

Observation: Let $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ and $\vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$. Note that we may express the dot product

$$\vec{x} \cdot \vec{y} = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \vec{x}^{\,t} \vec{y}.$$

Consequently, the (i, j)-entry of the matrix product AB may be expressed as

$$\begin{aligned}(i,j)\text{-entry of } AB &= (\text{row } i \text{ of } A) \cdot (\text{column } j \text{ of } B) \\ &= (\text{row } i \text{ of } A)^t (\text{column } j \text{ of } B)\end{aligned}$$

In other words, if $r_1, r_2, \ldots, r_m$ denote the rows of A and $c_1, c_2, \ldots, c_n$ denote the columns of B, then we may express

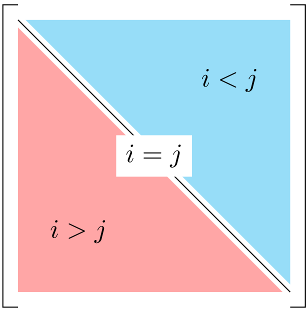
$$AB = \begin{bmatrix} r_1^t c_1 & r_1^t c_2 & \cdots & r_1^t c_n \\ r_2^t c_1 & r_2^t c_2 & \cdots & r_2^t c_n \\ \vdots & \vdots & \ddots & \vdots \\ r_m^t c_1 & r_m^t c_2 & \cdots & r_m^t c_n \end{bmatrix}.$$

We finally consider matrices which has special forms.

Definition 53

A square matrix $A = [a_{ij}]$ is

1. **upper triangular** if $a_{ij} = 0$ for $i > j$.
2. **lower triangular** if $a_{ij} = 0$ for $i < j$.
3. **diagonal** if $a_{ij} = 0$ for $i \neq j$.
4. **symmetric** if $A^t = A$.
5. **skew symmetric** if $A^t = -A$.



The entries a_{ii}, $i = 1, 2, \ldots, n$ of a square matrix $A = [a_{ij}] \in M_n(\mathbb{R})$ are said to be in the **main diagonal**. Thus, a diagonal matrix is a matrix with entries 0's off the main diagonal.

Example 54. The matrix

$$\begin{bmatrix} 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}$$

is upper triangular. The matrix

$$\begin{bmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -2 & 3 \end{bmatrix}$$

is lower triangular. The matrix below is a diagonal matrix

$$\begin{bmatrix} 9 & 0 \\ 0 & -1 \end{bmatrix}.$$

Since a diagonal matrix is completely determined by the entries in its main diagonal, we simply write $\text{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$ to denote the n-by-n diagonal matrix

$$\begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}.$$

It is easy to check that if $D_1 = \text{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$ and $D_2 = \text{diag}(\mu_1, \mu_2, \ldots, \mu_n)$ and a and b are scalars, then

$$aD_1 + bD_2 = \text{diag}(a\lambda_1 + b\mu_1, a\lambda_2 + b\mu_2, \ldots, a\lambda_n + b\mu_n),$$

and

$$D_1 D_2 = \text{diag}(\lambda_1\mu_1, \lambda_2\mu_2, \ldots, \lambda_2\mu_2).$$

Example 55. The matrix

$$S = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -2 & 4 \\ 3 & 4 & 10 \end{bmatrix}$$

is symmetric while the matrix

$$T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

is skew-symmetric.

Theorem 56

A square matrix $A = [a_{ij}]$ is symmetric if and only if $a_{ij} = a_{ji}$ for all i and for all j. On the other hand, A is skew-symmetric if and only if $a_{ij} = -a_{ji}$

The theorem above follows directly from the definition. We can say something more about a skew-symmetric matrix.

Corollary 57

If $A = [a_{ij}]$ is skew-symmetric, then $a_{ii} = 0$ for all i.

Proof. If $A = [a_{ij}]$ is skew-symmetric, then for all i and for all j, $a_{ij} = -a_{ji}$. Thus, in particular, $a_{ii} = -a_{ii}$ for all i. Hence, $2a_{ii} = 0$. Therefore, $a_{ii} = 0$ for all i. ▨

Corollary 57 implies that if a square matrix is skew-symmetric, then all its diagonal entries are zeroes.

